

CEL: Comprehensive Counterfactual Explanations Library and Benchmark

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What is a counterfactual explanation?

A small input change that flips a model's prediction.

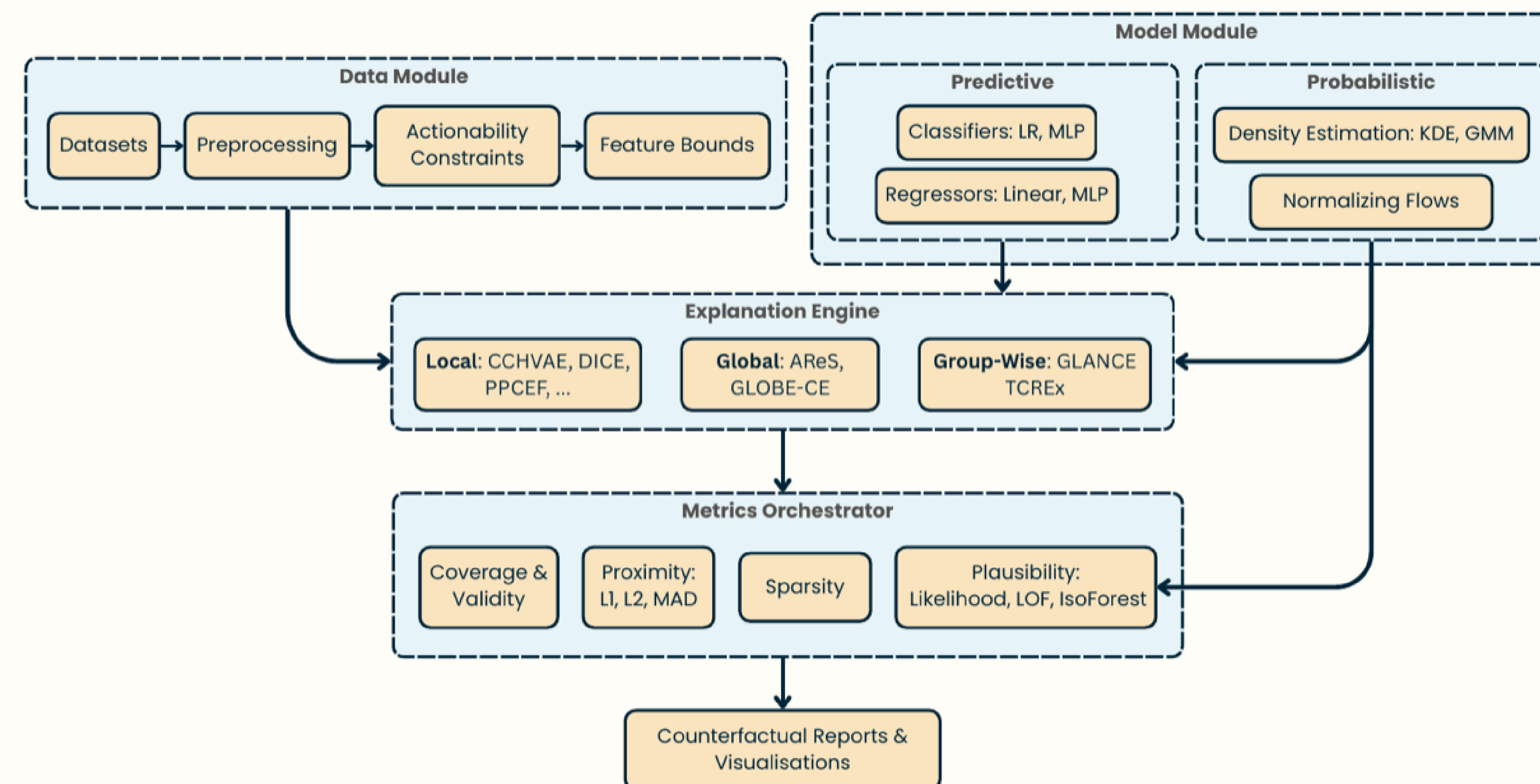
LOAN APPLICATION EXAMPLE

Feature	Original	Counterfactual
Age	35 years	35 years
Employment	Full-time	Full-time
Credit history	6 years	6 years
Loan amount	€15,000	€15,000
Monthly income	€3,000	€3,500
Monthly debt	€1,200	€700

Prediction **Declined** **Approved**

Only 2 of 6 features change.

One evaluation framework



Benchmark scope

18

datasets

14

methods

3

paradigms

2

backbones / task

5

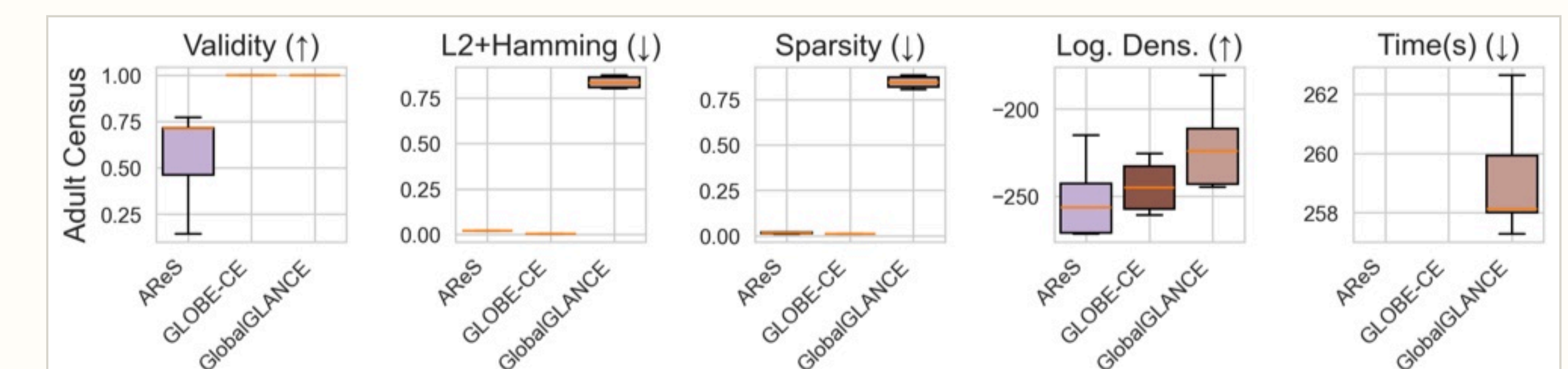
folds

9

classification metrics

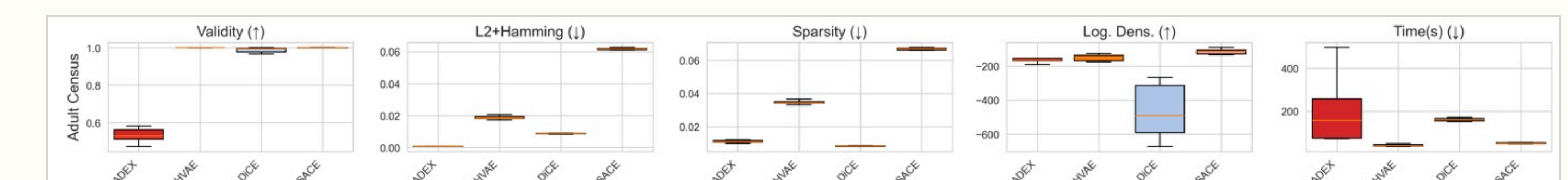
Results

Global: validity and failures



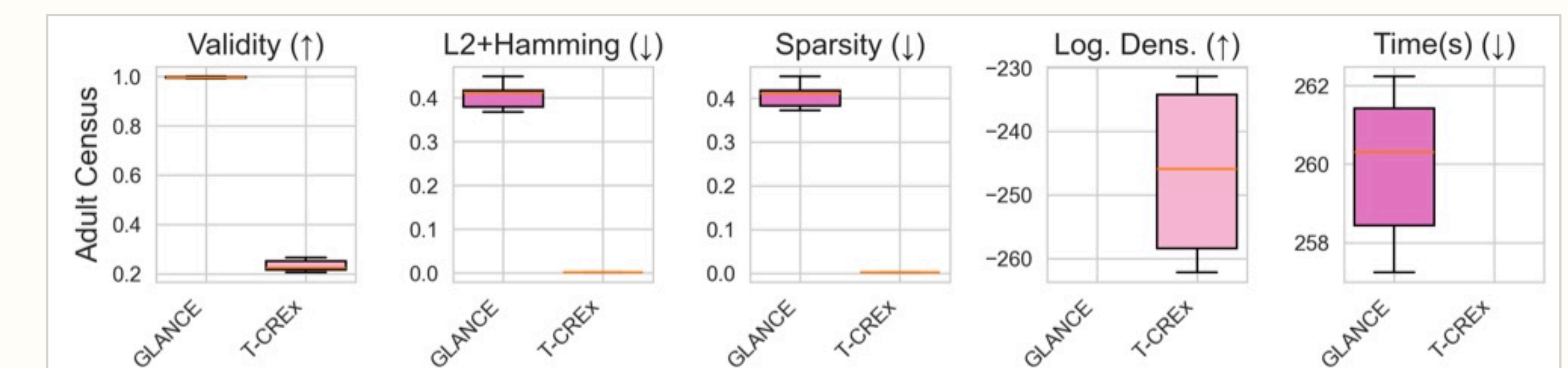
ADULT CENSUS · GLOBAL METHODS

Local: no single winner



ADULT CENSUS · LOCAL METHODS

Group-wise: reach versus change



ADULT CENSUS · GROUP-WISE METHODS

Three contributions

CONTROL Controlled evaluation protocol

COVER Broad benchmark across CE paradigms

EXTEND Extensible open-source benchmark workbench



Code & project